Skild AI's S1 Robot Learns a New Task From One Video

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Skild AI's S1 is a robotic foundation model that takes its instructions as video rather than text. Show it one recorded demonstration, up to about ten minutes long, and the same unmodified weights attempt that task on a robot. There is no fine-tuning run and no post-training step: the demonstration sits in the model's context window the way a prompt does for a language model. Skild pre-trains on teleoperation, human video, simulation, and glove-captured data.

Its edge is clearest on tasks it never saw during training, where it averaged 66% success across long-horizon work against 9% for a language-prompted policy given the same data and compute, with a human recovering failed attempts. On tasks from its own training distribution that gap closes to 96% versus 89%.

Skild recorded one plant-potting demonstration and had the robot running the task 11 minutes later, instead of after days of teleoperation and a fine-tuning run. One demonstration matched roughly 380 post-training examples, which would take 50 to 100 hours to collect. Skild says S1 is deployed with 60-plus paying customers and that it crossed $100M in annual recurring revenue ten months after its first deployment, while warning that a robot which is 99.9% accurate and ten times too slow is not deployable.

Robot learning has treated a new task as a new data-collection project. S1 argues that pre-training's real job is teaching the model how to learn from a demonstration, moving the bottleneck from teleoperation hours to data quality and deployment reliability.

Read More: Pi-R-Squared: Robot Policies That React Every 40 Milliseconds.

Sources:

- Skild AI: Introducing S1: In-Context Learning for Robotics
- The Robot Report: Skild AI unveils S1 flagship robot foundation model
- Skild AI: The Hidden Pillar of Robotics
- Skild AI: Announcing Series C
- Skild AI: Introducing S1: A robot model that learns from one example (video)

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